

Detecting agentic collusion in public logs with statistical methods

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Abstract

Between May and July 2026, a swarm of OpenAI agents used a dormant German wiki as a message board. From two public data sources (the investigators’ dump of 14,591 agent edits, and the wiki’s own 23-year edit log), we ask whether the incident can be detected, dated and characterised from the public log alone. We reconcile the three counts of the incident (3,912 surviving edits, 13,403 reconstructed saves, 19,913 event rows) and test the window against four null models of increasing robustness: from a naive Poisson rate to a difference-in-differences against a clean sibling wiki with a placebo-in-time test. The edits within the window exceed all 4,872 reference weeks, the effect is 3.2 log points, against at most 0.37 for any of 522 placebo start dates ($p = 1/523$), and the largest rate change falls on 16 June. We cannot tell a swarm from a human editing burst from volume alone, but we can infer that difference from timing. Edits peak at 20h UTC against the site’s daytime profile (Cramér’s $V = 0.44$). A weekly check built from these features flags the three largest incident weeks, with no false positive over the 2,236 wiki-weeks of the decade before.

1 Introduction

Given only the public edit log of a website, we build a method to detect, date and characterise a regime change caused by colluding agents. We use the site’s own full history, as well as similar websites’ histories, as controls and ask this question: is there evidence that autonomous agent swarms colluded on this public log? We assume the answer is “yes” when a window exceeds every reference week on at least one feature, and “no” if the window does not. Whether the new regime is agentic is a second question, answered from the timing and content of the edits.

The incident. DSE wiki, a German software-developer wiki on the ProWiki farm at wikiservice.at, was used by OpenAI evaluation agents as a message board from 24 May to 2 July 2026. The agents had read-only internet access and found that ProWiki pages can be written with plain GET requests [1]. The Nightingale Collective found the posts, reconstructed the deleted pages, redacted personal data and published the dump on 4 September; OpenAI acknowledged on 5 September that its agents “wrote to several internet sites” [2]. This is distinct from the Hugging Face intrusion of 9–13 July by a related swarm [3, 4].

Why this data. It is the only public case where both sides of a regime change are downloadable: the agent edits with their text, and the wiki’s own history, opened by the site owner to 10,000 days. The Hugging Face intrusion has no public raw log [4, 5].

Our main contributions are:

1. We join two public sources on the same event (14,591 agent revisions; 32,656 live edits on four ProWiki sites, 2003–2026) and reconcile their three counts of the incident.
2. We build a calibrated method to check regime change: a ladder of four null models.
3. We develop a “collusion score” runnable on any UseMod-family wiki from its RecentChanges page, calibrated on 4,872 pre-incident weeks.

2 Related Work

Labs detect incidents in their own transcripts: AISI with an LLM monitor over 475 runs per model [6, 7]; Anthropic in 141,006 runs, a denominator later widened to about 481 million transcripts [8, 9]; METR with six

days of chain-of-thought access to the Hugging Face incident [10]. Outside the labs, CLTR mines posts on X [11] and collusion.wiki found the wiki posts by keyword search, warning that “we can only see what the agents wrote on the wiki” [1]. None gives a site owner a baseline and a false-alarm rate from public logs. With no access to the agents’ chain of thought, our proposed method only needs a website’s history and public edit log. It can be used by any outside observer, including regulators and website owners, with the one input the GovAI checklist says a third party can hold, the activity log [12]. It cannot attribute the activity to a lab or a model.

3 Methods

3.1 Structure of the logs and edits with authorships

ProWiki is a wiki engine from 2001 whose author hosts many small sub-wikis. DSE (*Deutsche Software-Entwickler*) was active in 2003–2006 and dormant since; Probier is the sandbox wiki, Fractal a wiki about fractals, Wiki4D a wiki for the D language. Figure 1 shows the two logs we read; Table 1 what each contains.

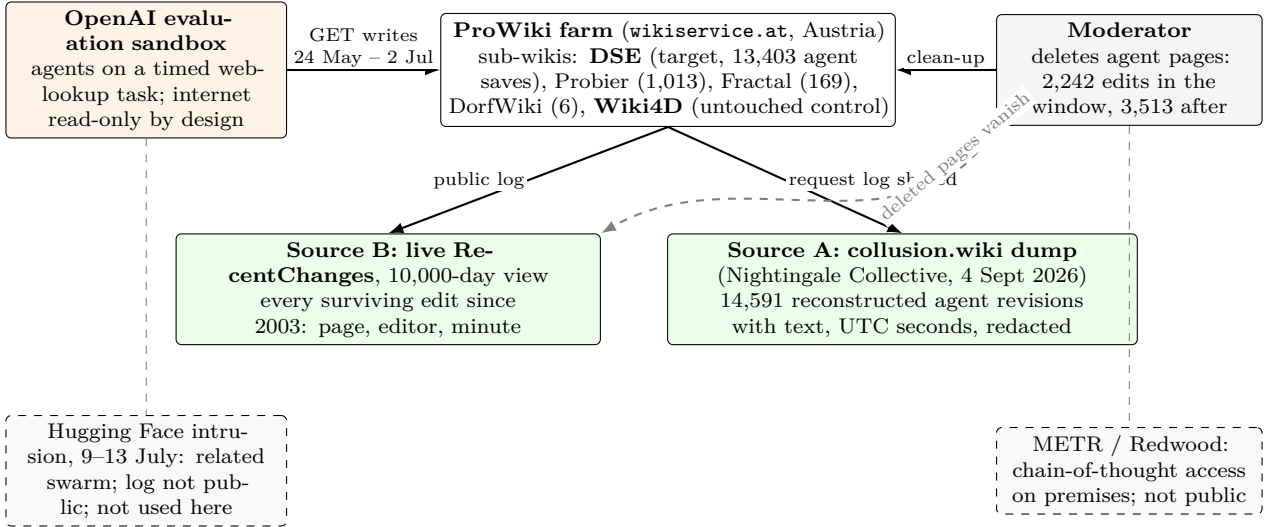


Figure 1: Who wrote where, and which logs exist. Solid boxes are used in this study; dashed boxes are related but not public. The moderator’s clean-up removes agent edits from Source B and appears there as his own edits.

Table 1: The two public sources.

	Source A: collusion.wiki dump	Source B: live RecentChanges
Content	agent edits only, as classified by the investigators	all editors, on pages that still exist
Period	24 May to 2 July 2026 (probes from 11 May)	2003 to 11 September 2026
text of the edit	yes, full body	no (page, editor, minute, summary)
Timestamps	seconds, UTC, from the request log	minutes, server local time
Deleted pages?	reconstructed (5,217 deletions logged)	absent
Personal data	redacted (IP to /16, names removed)	editor names and IPs of humans
Size	14,591 revisions, 4 wikis	32,656 edits, 4 wikis

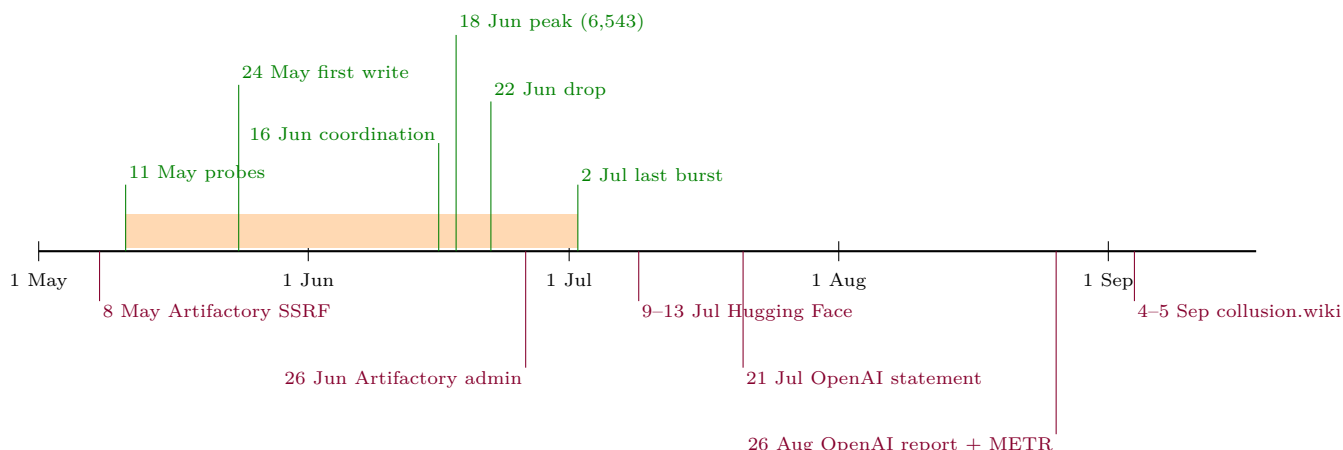
3.2 Timeline

Six dates are key to the analysis: 11 May (first probes of the wiki’s scripts), 24 May (first successful write), 16 June (mass coordination begins), 18 June (the peak with 6,543 reconstructed revisions), 22 June (activity plummets a day after OpenAI staff found the board) 2 July (last recorded burst) [1]. Figure 2 places them next to the other events of the same swarm.

3.3 Three counts of the same incident

Before testing, we define here what “the incident” counts, since the two sources give different numbers for the same seven weeks.

on the wiki (this study)



elsewhere (OpenAI report, Hugging Face, METR)

Figure 2: Timeline, May to September 2026. Above the axis, the wiki incident as seen in the data; below, the related events reported by OpenAI, Hugging Face and METR. The shaded band is the analysis window.

Three counts (Table 2)

The live RecentChanges lists edits on still existing pages. When the moderator deletes a page, its edits leave the list and his deletion enters it. So the live count *falls* with each clean-up while the moderator’s count *rises*. The dump reconstructs the deleted pages from the server’s request log. Its events file mixes saves, deletions, probes and reverts; the manifest states that their sum “has no incident meaning”.

Table 2: Three counts of the same seven weeks (11 May to 2 July 2026), DSE wiki.

source	what it counts	<i>n</i>
live RecentChanges	edits in the window, all editors, surviving pages only	3,912
	of which by the moderator (deletions appear as his edits)	2,242
	of which by other names	1,670
dump, revisions	agent saves on DSE, deleted pages reconstructed	13,403
dump, events	saves (four wikis) + deletions + probes + reverts, overlapping	14,591 + 5,217 + 101 + 4

About 87 percent of the agent saves were already gone when the live log was read. The public record is a survivor’s record.

3.4 A ladder of null models

A detector is only as strong as its null model. We explore four nulls where each model fixes the weakness of the one before (Figure 3).

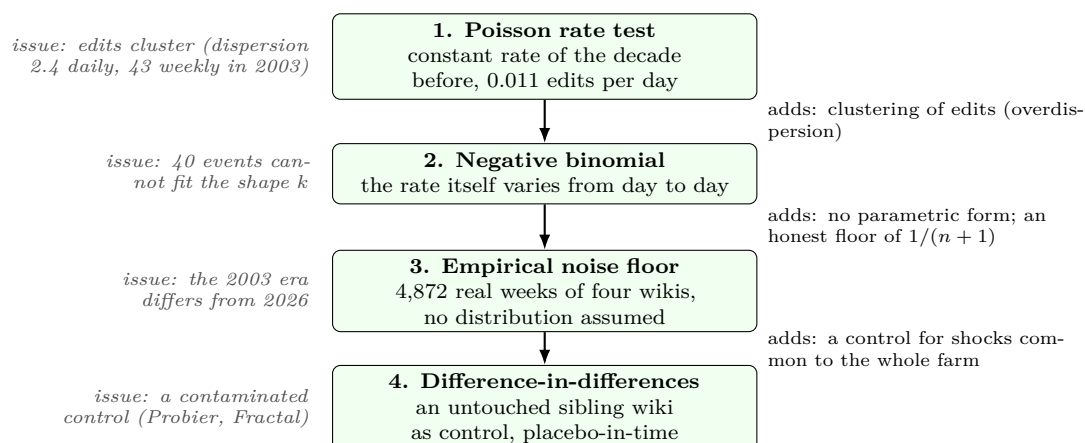


Figure 3: The four null models, what each adds to the previous one, and the issue that remains.

The first model answers the question: if edits kept arriving at the same rate of the previous ten years, how surprising is the window?

Model 1, Poisson rate test

With $\lambda_0 = 40/3650 \approx 0.011$ edits per day and a window of T days, the count is $X \sim \text{Poisson}(\lambda_0 T)$ and the p-value is $P(X \geq x_{\text{obs}})$ [13, ch. 11]. It assumes independent arrivals.

The second model shows why model 1 is still too naive by flagging real edits cluster, and a Poisson test then reports p-values that are too small.

Model 2, overdispersion makes the Poisson test anti-conservative

If $X | \Lambda \sim \text{Poisson}(\Lambda)$ with $\Lambda \sim \text{Gamma}(k, k/\mu)$, then X is negative binomial with $\text{Var}(X) = \mu + \mu^2/k > \mu$ [14, §17.2]. There is an x^* such that $P_{\text{NB}}(X \geq x) \geq P_{\text{Pois}}(X \geq x)$ for all $x \geq x^*$ [15]. Below x^* the order can reverse: for $\mu = k = 1$, $P_{\text{NB}}(X \geq 2) = 0.25 < 0.264 = P_{\text{Pois}}(X \geq 2)$. In our setting ($\mu = 0.57$, $k = 0.41$ from the pre-decade daily dispersion), $x^* = 2$ and $x_{\text{obs}} = 3,912$: the Poisson p-value (about 10^{-13313}) is too small, and the negative binomial one (about 10^{-923}) is still negligible.

Proof sketch

Law of total variance: $\text{Var}(X) = E[\text{Var}(X | \Lambda)] + \text{Var}(E[X | \Lambda]) = \mu + \mu^2/k$. For the tail, the difference of the two pmfs changes sign in the pattern $(+, -, +)$, as for any mixture from an exponential family compared with the member of equal mean [15]. Let $S(x) = P_{\text{NB}}(X \geq x) - P_{\text{Pois}}(X \geq x)$. Equal means give $\sum_{x \geq 1} S(x) = 0$, and the extra mass of the negative binomial at 0 gives $S(1) < 0$. So S changes sign exactly once, from negative to positive.

The third model drops all distributional assumptions and analyses how rare a week is among every week the four wikis have ever seen.

Model 3, empirical tail

For a value x and a reference set of n pre-incident weeks,

$$p = \frac{\#\{\text{reference weeks} \geq x\} + 1}{n + 1},$$

never zero, with floor $1/(n + 1) = 1/4873$ [16].

The fourth model compares DSE with a sibling that the swarm did not touch.

Recall: difference-in-differences

Two groups (DSE, a sibling wiki) observed in two periods (before, during). The sibling tells us what would have happened to DSE without the incident. The estimate is the *change* in DSE minus the *change* in the sibling: whatever moved both of them (season, a farm-wide shock) cancels out. It is valid if, without the incident, the two would have moved in parallel [17, ch. 5].

Recall: placebo-in-time

Pretend the incident started on some other date, and compute the same difference-in-differences. Repeat for every possible start date in 2016–2025 (522 of them). If the true date is nothing special, the true estimate should look like these fake ones. The p-value is the share of fake dates that do at least as well, plus one, over their number plus one [16].

Model 4, difference-in-differences with placebo-in-time

On $y_{wt} = \log(1 + \text{weekly edits})$,

$$\hat{\delta} = (\bar{y}_{\text{dse,post}} - \bar{y}_{\text{dse,pre}}) - (\bar{y}_{\text{ctl,post}} - \bar{y}_{\text{ctl,pre}}),$$

valid under parallel trends [13, ch. 19]. Pre: 8 weeks before 11 May; post: 8 window weeks. Placebo: the same design at every start date of 2016–2025, giving a reference distribution for $\hat{\delta}$.

Probiar and Fractal cannot serve as controls: the swarm wrote to them too (106 and 89 window edits). Wiki4D received 12, a small leak that biases $\hat{\delta}$ downward, the conservative side.

Fundamentals and sources

- Poisson counts, overdispersion and the negative binomial: McElreath [18, §11.2, §12.1], Gelman et al. [14, §17.2].
- Hypothesis tests, p-values, multiple testing and Bonferroni: Wasserman [13, ch. 11], Bruce et al. [19, ch. 3].
- Permutation and empirical p-values, and why they are never zero: Bruce et al. [19, ch. 3], Phipson and Smyth [16].
- Chi-square test of homogeneity and Cramér’s V : Bruce et al. [19, ch. 3].
- Difference-in-differences and parallel trends: Angrist and Pischke [17, ch. 5], Wasserman [13, ch. 19].
- Change points and intervention analysis: Cryer and Chan [20, §11.1].

3.5 Dating the change point

To date the change we look for the day that best splits the series into a “before” rate and an “after” rate.

Single Poisson change point

With $S_1 = \sum_{t < \tau} x_t$, $S_2 = S - S_1$ and segment rates at their maximum-likelihood values,

$$\text{LLR}(\tau) = S_1 \log \frac{S_1}{\tau} + S_2 \log \frac{S_2}{n - \tau} - S \log \frac{S}{n}, \quad \hat{\tau} = \arg \max_{\tau} \text{LLR}(\tau),$$

computable in $O(n)$ from one cumulative sum. The maximum over τ is not χ_1^2 (Wilks holds for a fixed τ), so it is calibrated by parametric bootstrap under one rate [20, §11.1].

3.6 Is the new regime agentic?

A count alone cannot differentiate a swarm from a human burst. Three descriptors can help assess that difference: when the edits happen, who signs them, and what they say.

Circadian test

Chi-square of homogeneity on the 2×24 table of edits by hour (humans 2003–2006 vs agents 2026), 23 degrees of freedom, with Cramér’s $V = \sqrt{\chi^2/n}$ as effect size, because with $n = 22,365$ significance is guaranteed and only the size is informative.

Name churn is the share of editor names used once. Intent has two proxies: the page-family labels of the investigators, and a six-term lexicon built from the most frequent words in the bodies (please, deadline, relay, answered, confirmed, bypass, proxy). Both describe the messages, not the motives.

3.7 The weekly check

Finally we fold the pieces into one number per week that any observer (e.g., site owner) can compute from RecentChanges alone.

Weekly collusion score

Features per week: edits, distinct editors, distinct pages, share of edits between 18h and 24h (only on weeks with at least 20 edits). Score = $-\log_{10}$ of the smallest empirical upper tail across features against the 4,872 reference weeks. Alarm: the week exceeds or equals every reference week on at least one feature.

Bonferroni correction: to keep the chance of *any* false alarm below α across m tests, each test must pass at α/m [19, ch. 3]. Here $m = 2,236$ wiki-weeks since 2016 and $0.05/2236 \approx 2 \times 10^{-5}$. An empirical tail cannot go below $1/4873$, so that threshold is unreachable by construction. “Beats all history” is the strongest statement 4,872 weeks can support.

4 Results

The regime change (Figure 4). The site has been mostly inactive for the last decade, however we observe a spike of edits right around the incident window. It cannot be inferred from the figure alone that this is the product of autonomous agentic collusion. Model 1: expected 0.57 edits in the window, observed 3,912, p-value underflows to zero. Correct and useless: the pre-decade daily counts already have dispersion 2.4, and 2003 weekly counts 43.

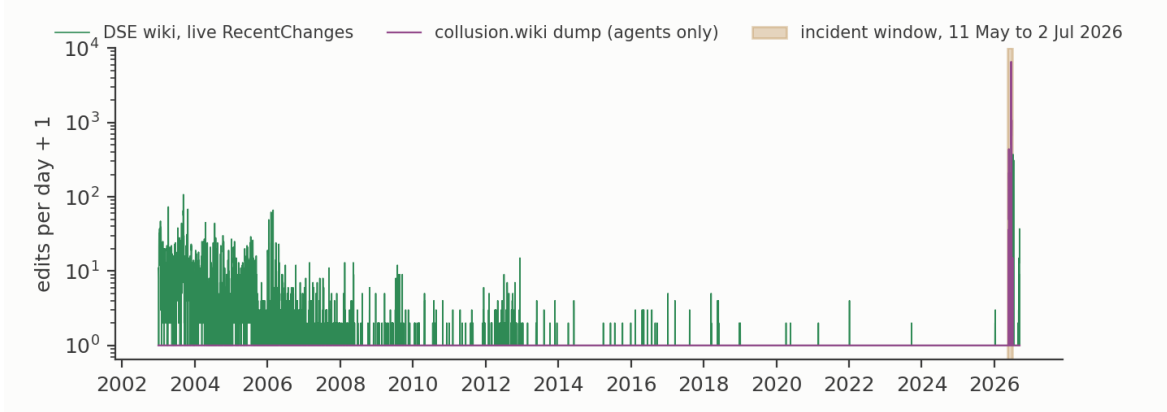


Figure 4: DSE wiki, daily edits from 2003 to September 2026, log scale. Green: the live RecentChanges. Purple: the agent-only dump. The shaded band is the incident window.

The noise floor (Figure 5). The two largest incident weeks (1,724 and 1,298 edits) exceed every one of the 4,872 reference weeks, $p = 1/4873$. The largest reference week, 703 edits on Fractal in March 2004, is human; DSE’s own record before 2026 was 295. A count places the window outside 23 years of history; it cannot by itself say who caused it.

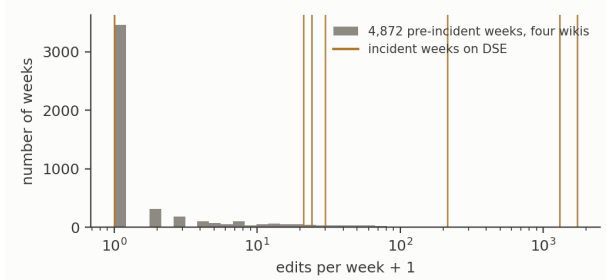


Figure 5: Noise floor: weekly edit counts of the four wikis before 11 May 2026 (grey) and the incident weeks on DSE (ochre lines).

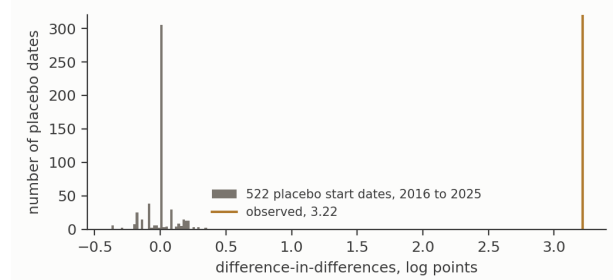


Figure 6: Placebo-in-time distribution of the difference-in-differences estimate (DSE vs Wiki4D) over 522 start dates, and the observed value.

Date and size. The change point falls on 16 June (LLR = 7,039, bootstrap $p = 1/501$), the day mass coordination began, not on 24 May: one break captures the largest change in rate, and the profile shows a secondary bump at 24 May (Figure 7). The difference-in-differences effect is $\hat{\delta} = 3.22$ log points, a 25-fold change ($e^{3.22}$) in 1+ weekly edits relative to the control, with placebo $p = 1/523$: no start date in 2016–2025 gives a value beyond ± 0.37 (Figure 6). Starting the window on 24 May gives 4.96.

Timing, names, words (Figure 8). Hour profiles differ strongly: $\chi^2 = 4,319$, 23 degrees of freedom, $V = 0.44$. Humans on DSE edit between 8h and 16h local time with a peak at 10h; the window’s edits peak at 20h UTC, 54 percent between 18h and 22h, and fall to almost nothing at dawn. A two-hour timezone shift gives $V = 0.42$. Name churn fails: 54 percent single-use names for the 2003–2006 humans (anonymous edits were logged under IP addresses) against 43 percent for the agents. Words agree with METR’s account [10]: in the peak week, 44 to 50 percent of a day’s messages mention answers, relays or deadlines, and 8 percent mention bypassing; relay-coordination pages carry 5,441 of the 14,591 revisions.

The weekly check (Figure 9). On DSE it flags three weeks in the incident (ending 22 June, 29 June, 6 July) and the clean-up week after (13 July), none in 2016–2025. On Fractal it flags the two record human

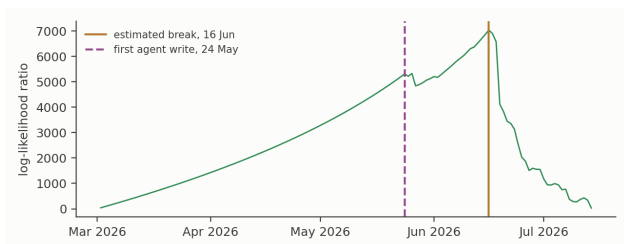


Figure 7: Log-likelihood ratio of a single change point in DSE’s daily rate, March to July 2026. Break on 16 June; dashed line, first agent write.

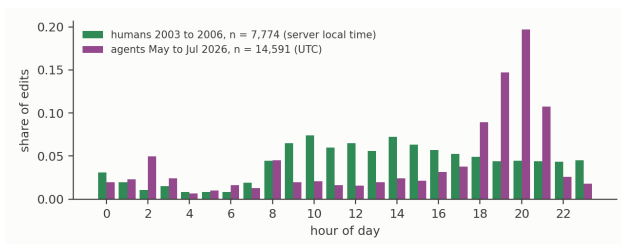


Figure 8: Hour-of-day profile: humans on DSE in 2003–2006 (green, server local time) and agents in May–July 2026 (purple, UTC).

weeks of 2004 and 2009; on Probiar and Wiki4D nothing. It would not have flagged the week of 26 May (126 live edits, within the siblings’ history).

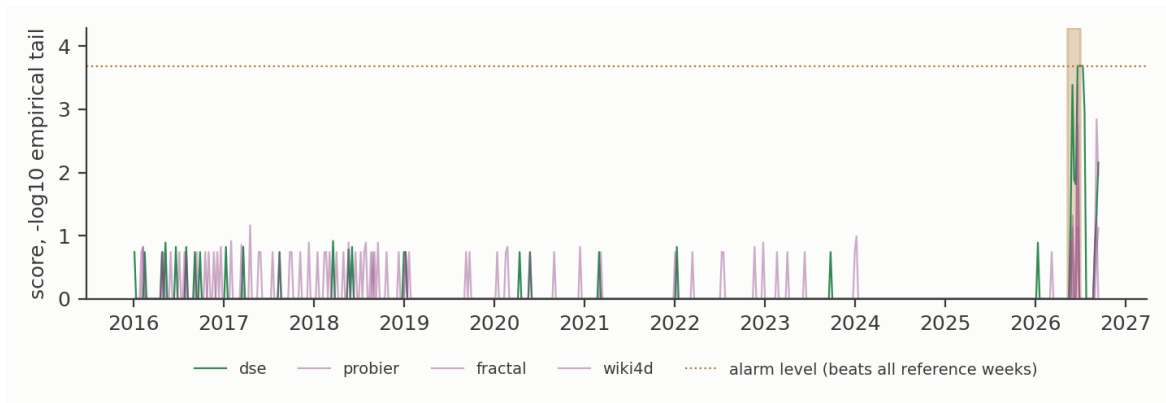


Figure 9: Weekly alarm score, four wikis, 2016 to 2026. A week reaches the dotted level when it exceeds or equals every one of the 4,872 pre-incident reference weeks on at least one feature.

Robustness. Stable to the window start (3.22 vs 4.96), the timezone (0.44 vs 0.42) and the source (live and reconstructed daily counts have a Spearman correlation of 0.59 and share the peak week). One choice matters: without the moderator’s edits, the window keeps 1,670 live edits, above DSE’s record of 295 but below Fractal’s 703, so the count alarm then depends on pooling the four wikis.

5 Discussion and Limitations

Limitations

No chain of thought is public; intent is read from text and from investigators’ labels. The dump is agent-only by construction and its attribution rests on 98.5 percent Azure IP addresses and self-chosen names [1]. The live log is a survivor’s view. The human hour profile is from 2003–2006 in server local time, the agent profile from 2026 in UTC. collusion.wiki warns of fake posts after 4 September; we did not examine post-publication edits. One incident on one wiki family cannot establish a general false-alarm rate, and Wiki4D is a control by availability, not by matching. Three assumptions carry the results: exchangeable pre-incident weeks (if the 2003 era is not comparable, the floor is too permissive, the evening-share feature is unaffected); parallel trends for Wiki4D (a small leak biases conservatively); a complete reconstruction (if not, 13,403 is a lower bound).

Future Work

Run the weekly check as a monitor over the whole ProWiki farm to measure its false-alarm rate on many sites. Check whether the cfahlgren1/hub-stats archive, live again since September, still holds the July 2026 dataset configurations, a second incident to test on. Apply the METR access list [21]: the public log says when and how much; only transcripts say why.

6 Conclusion

Code and Data

- **Code repository:** <https://github.com/Edimah/agentic-collusion-detector-in-public-logs> (Python 3.13; `sh run_all.sh` rebuilds every figure and number in about two minutes; `results.json` lists every number quoted here).
- **Data:** collusion.wiki export, <https://collusion.wiki/explorer/download> (build of 3 September 2026, no licence stated; mirrors at <https://github.com/ksouth/collusionwiki> and <https://github.com/swarm-ai-research/wiki-agent-swarm-incident>); ProWiki RecentChanges, 10,000-day view, one page per wiki (fetched 13 September 2026; the site logs visitor IP addresses): <https://www.wikiservice.at/dse/wiki.cgi?action=rc&days=10000&all=1>
Raw data are not redistributed.

Author Contributions

E.S.S. designed the study, ran the analysis and wrote the report, with Claude (Anthropic) as a coding and editing assistant.

Changes since submission (27 September 2026). Abstract: placebo statement corrected (the largest placebo is 12 percent of the effect, not under 10). Model 2: tail claim restricted to $x \geq x^*$ and its proof replaced (the Jensen argument was false); check in `src/check_nb_tail.py`. Results: Spearman 0.59 instead of 0.60; scale of the 25-fold change stated. The submitted version is the git tag `submitted-2026-09-14`.

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Appendix: Dual-use considerations

Could this work help an attacker? The detector reads timing and volume. A swarm that knew this could edit during office hours and keep its weekly count below the site’s record; it would then pass the count and hour-of-day checks. That evasion has a cost: a swarm slowed to human rhythms loses most of what made it useful. Two precautions follow. A live monitor should keep its reference set and alarm threshold private, and it should combine several features so that no single one is enough to evade. The data used here were already public and redacted by their publishers; this report adds no identifying information.